

Routing-Informed Sparse ETM for Neural MS2LDA: Validation Checkpoint and Model Selection

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Abstract

Mass2Motif discovery requires more than good held-out likelihood. A useful model must retain a broad, non-redundant topic inventory and assign short tandem mass spectra strongly enough for chemical evaluation. Canonical and balanced Embedded Topic Models (ETMs) retained reasonable likelihood but produced diffuse document mixtures, while a sparse ETM using entmax alone starved most topics.

We therefore retained the published ETM generator and added two bounded inference adaptations: top-2 contextual evidence from observed spectral words and published alpha-entmax 1.5. The resulting model has only one additional learned scalar. Across three unchanged training seeds on the locked seed-42, $K = 1000$ MSⁿLib validation split, it produces 439–453 evaluable and 274–289 useful motifs, compared with 408/265 for the private M1 reference and 206/138 for Tomotopy. Median effective topics remain 3.70–3.71, 813–828 topics win at least one validation spectrum, and no catastrophic duplicate component appears.

The model does not dominate every metric. Across the same runs it produces 787–803 optimized motifs, mean SOS 0.637558–0.647350 and completion NLL 9.539388–9.546012, versus 884, 0.658079 and 8.974140 for M1. Three conservative M1-relative gates are therefore consistently false and candidate test remains locked. The correct scientific conclusion is a strong near-pass and viable paper-facing baseline: the discovery breadth, sparsity and inventory repairs are initialization-stable on this split, while residual chemical-quality and likelihood trade-offs also reproduce. A final weight-1 positive-NPMI add-on improved its own train-graph objective but reduced planted beta recovery, so it stopped at the first synthetic gate and was never run on real validation. The last plausible zero-parameter simplification, direct top-2 token voting without spectral context, also failed its first gate: beta/theta recovery fell by 0.0881/0.1034. This isolates the one learned context scalar as necessary in the simplest demonstrated formulation.

1 Scientific objective

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss tokens as words, and recurrent co-fragmentation patterns as Mass2Motifs [1]. At $K = 1000$, a useful model must simultaneously:

1. predict held-out spectral evidence;
2. retain many distinct topic-word distributions;
3. assign each short spectrum to only a small number of motifs;
4. produce enough confident spectrum-topic associations for MAG/SOS; and
5. retain chemically useful motifs under leakage control.

Likelihood alone can hide major failures. Topics may be starved, duplicated or nearly uniform, and a spectrum may spread probability over dozens of topics even when completion NLL is good. Earlier ETM experiments reproduced both failure classes. This study asks whether a recognizable ETM base can be adapted to the short MS/MS document regime without rebuilding the private M1 system.

2 Evidence and metric contract

The experiment uses the positive-mode MSⁿLib assets from Zenodo record 20179680 [3, 7]. The locked scaffold/compound-disjoint split contains 27,222 training and 3,889 validation spectra. Vocabulary and 48-dimensional SGNS features are constructed from training spectra only. Physical peak groups are divided as units for document completion, preventing a fragment and its corresponding neutral loss from leaking across observed and withheld halves.

MAG retrieves leakage-filtered reference spectra and optimizes motif annotations. The frozen validation membership threshold is $\theta_{dk} \geq 0.5$. The primary metrics are:

- **optimized motifs**: topics for which MAG produces at least one optimized annotation feature;
- **evaluable motifs**: optimized topics associated with at least one unique validation molecule;
- **SOS**: annotation-fingerprint containment in a molecule fingerprint, averaged over unique molecules and then over topics;

- **useful motifs**: evaluable topics with $\text{SOS} \geq 0.6$; and
 - **completion NLL**: negative log-likelihood per in-vocabulary withheld token under $\theta\beta$, where lower is better.
- Candidate selection used training plus validation only. Candidate test theta, completion, MAG, SOS and result files were not loaded, calculated or inspected. The committed access audit records this boundary explicitly.

3 Routing-informed sparse ETM

3.1 Published ETM generator

Let $\rho_w \in \mathbb{R}^{48}$ be the fixed train-only SGNS coordinate of word w and $\alpha_k \in \mathbb{R}^{48}$ the learned ETM coordinate of topic k . The decoder is the ordinary embedding-space topic model [4], with separate fragment and neutral-loss normalizations so that each channel receives half of every topic’s probability mass:

$$\beta_{kw} = \begin{cases} \frac{1}{2} \text{softmax}_{v \in V_F}(\alpha_k^\top \rho_v), & w \in V_F, \\ \frac{1}{2} \text{softmax}_{v \in V_L}(\alpha_k^\top \rho_v), & w \in V_L. \end{cases} \quad (1)$$

The two-layer ETM encoder returns Gaussian parameters $\mu(x)$ and $\sigma(x)$ for normalized spectrum bag-of-words x . The variational family, standard-normal analytic KL, raw pseudo-count multinomial reconstruction and optimizer remain unchanged.

3.2 Top-2 contextual posterior evidence

For each non-zero word, the model combines its normalized SGNS vector with the weighted leave-one-out mean of the other observed words. One learned scalar c controls the context contribution:

$$h_{dw} = \text{normalize} \left(\text{normalize}(\rho_w) + c \frac{\sum_v x_{dv} \rho_v - x_{dw} \text{normalize}(\rho_w)}{\max(1 - x_{dw}, \epsilon)} \right). \quad (2)$$

The contextual word is compared with the same normalized topic coordinates that define β :

$$s_{dwk} = \frac{h_{dw}^\top \text{normalize}(\alpha_k)}{\tau_r}, \quad \tau_r = 1.0. \quad (3)$$

Only its two largest topic logits are retained (the top-2 route) and normalized with a softmax. Their word-weighted sum yields one document evidence distribution r_d .

The ordinary ETM posterior mean receives a bounded row-centred log offset:

$$\tilde{\mu}(x_d) = \mu(x_d) + \text{center} \left[\log \left(r_d + \frac{1}{K} \right) \right], \quad (4)$$

$$z_d = \tilde{\mu}(x_d) + \sigma(x_d) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (5)$$

$$\theta_d = \text{entmax}_{1.5}(z_d), \quad (6)$$

$$p(w | d) = \sum_k \theta_{dk} \beta_{kw}. \quad (7)$$

The $1/K$ pseudocount bounds topics absent from the sparse route. Row centring makes uniform evidence an exact no-op. Alpha-entmax 1.5 supplies exact zeros [5]. Routing temperature and strength are fixed at 1.0. The learned context scalar finished at 0.1772, and total parameters increase from 19,278,000 for balanced ETM to 19,278,001.

There is no nonlinear M1 router, document gate, Sinkhorn target, NPMI loss, prototype-separation loss, alternating optimizer or temperature schedule. The paper-facing claim is therefore ETM plus one explainable short-spectrum posterior adaptation and a published sparse simplex mapping.

4 Synthetic mechanism isolation

Truth-known synthetic spectra used 18 planted motifs, 1–3 generating motifs per spectrum, paired fragment/loss words, train-only SGNS, raw intensity pseudo-counts, 800 training and 160 validation spectra, and frozen seeds 11, 23 and 37.

Entmax alone makes mixtures sparse but worsens topic starvation. Routing alone improves recovery and breadth but remains too diffuse in a separate deliberately diffuse stress condition. Their combination is complementary: routing supplies a broad recoverable posterior signal, and entmax converts it into sparse mixtures.

At seed-11 $K=128$, the selected combination recovered all 18 planted motifs as distinct top-1 topics. It achieved beta/theta recovery 0.6638/0.9556, top-motif accuracy 0.9313, median 1.45 effective topics, median exact support 2 and NLL 6.1276. This high-overcompleteness result supported promotion of one fixed configuration to real validation.

Table 1: K=36 synthetic means. Beta and theta columns are cosine recovery against planted truth.

Formulation	NLL	Beta	Theta	Median eff.	Active	Unique top-1
Balanced ETM + softmax	6.4710	0.3052	0.4868	3.59	8.0	6.7
Balanced ETM + entmax	6.5977	0.2566	0.3973	1.88	5.0	4.7
Top-2 context + softmax	6.2438	0.4423	0.7066	3.22	13.3	11.7
Top-2 context + entmax	6.2892	0.4648	0.7513	2.00	14.0	13.0

5 Real MSⁿLib validation

5.1 Discovery breadth relative to M1 and Tomotopy

Table 2: Representative original run on the locked seed-42 validation split. Bold marks the best value in each row, not a global model winner.

Metric	M1	Routing ETM	Tomotopy
Optimized motifs	884	803	607
Evaluable motifs	408	445	206
Useful motifs	265	289	138
Mean SOS	0.658079	0.647153	0.676149
Median SOS	0.648864	0.657895	0.685450
Completion NLL ↓	8.974140	9.542924	9.662228

Routing ETM is better on discovery breadth: it adds 37 evaluable and 24 useful motifs over M1 and 239/151 over Tomotopy. It also has lower completion NLL than Tomotopy. It is not uniformly better: M1 retains higher optimized coverage, mean SOS and likelihood, while Tomotopy has higher SOS over a much smaller evaluable inventory.

5.2 Same-split training-seed stability

Two additional runs changed only model initialization and minibatch-order RNG. The data split, vocabulary, SGNS, model configuration, 120 epochs and chemical evaluation remained frozen. The original deterministic seed is 7043; the added seeds are 23 and 37.

Table 3: Real validation stability. The mean is descriptive for three initializations on one data split.

Seed	Optimized	Evaluable	Useful	Mean SOS	NLL	Median eff.	Unique
7043	803	445	289	0.647153	9.542924	3.699	828
23	791	453	274	0.637558	9.546012	3.702	816
37	787	439	275	0.647350	9.539388	3.714	813
Mean	793.67	445.67	279.33	0.644020	9.542775	3.705	819.0

Every run exceeds M1’s 408 evaluable / 265 useful motifs and Tomotopy’s 206/138. Every run also remains below M1 on optimized coverage and mean SOS and above M1 on completion NLL. Median exact support is 6 in every run, all runs have more than 800 unique validation winners, and none has a catastrophic duplicate component. The result therefore establishes initialization stability of both the discovery advantage and the residual trade-off on this frozen split. It does not establish data-split or external-dataset generalization.

The SOS bands explain the apparent mean/median tension.

Routing ETM increases the intermediate-quality inventory rather than reporting only a small high-scoring subset. Evaluable-to-optimized conversion is 55.4% versus M1’s 46.2%, and useful-to-optimized conversion is 36.0% versus 30.0%. Its median SOS is also higher than M1’s.

Table 4: Validation SOS composition. Useful motifs are the high and intermediate bands combined.

SOS band	M1	Routing ETM	Difference
High, > 0.8	79	67	-12
Intermediate, 0.6–0.8	186	222	+36
Low, < 0.6	143	156	+13
Evaluable total	408	445	+37
Useful total	265	289	+24

5.3 Sparse spectra without global collapse

Table 5: Diagnostics distinguishing per-spectrum sparsity from global topic collapse.

Routing ETM diagnostic	Value
Median / mean effective topics per spectrum	3.70 / 4.09
Median / p95 / maximum exact support	6 / 13 / 23
Unique top-1 topics	828
Topics active above mean usage 0.0005	535
Corpus-effective topics	538.40
Maximum mean topic usage	0.0143
Mean nearest-topic beta cosine	0.1529
Largest component at beta cosine ≥ 0.999	2 topics
Catastrophic duplicate component	no

Balanced softmax ETM used 46.72 median effective topics and produced only 166 evaluable / 104 useful motifs. Entmax alone used 1.80 effective topics but collapsed to 20 unique top-1 topics and 7 evaluable / 6 useful motifs. Routing plus entmax fixes both failures simultaneously: sparse document mixtures no longer arise by starving the global inventory.

5.4 Formal gates and scientific interpretation

The gates were frozen before scoring. Motif counts retain 95% of M1, mean SOS retains 99%, and NLL may be at most 105% of M1. Every gate had to be true for the formal pass.

Table 6: Frozen gate audit. The Boolean remains false even though the candidate is scientifically useful.

Gate	Threshold	Candidate	Distance	Result
Optimized motifs	≥ 840	803	-37	fail
Evaluable motifs	≥ 388	445	+57	pass
Useful motifs	≥ 252	289	+37	pass
Mean SOS	≥ 0.651498	0.647153	-0.004345	fail
Completion NLL	≤ 9.422847	9.542924	+0.120077	fail
Finite/stable	required	yes	—	pass
No catastrophic component	required	yes	—	pass

The mean-SOS miss is 0.67% below its threshold and NLL is 1.27% over its threshold. Optimized coverage is 4.40% below its gate. The all-gates value must remain false for audit integrity, but it is misleading to translate that Boolean into a claim that Routing ETM is generally worse than M1. The saved evidence supports the status *strong near-pass and current paper-facing baseline*.

6 Historical alternatives and negative evidence

The campaign tested one mechanism at a time and retained negative results:

- canonical fixed-SGNS ETM had good completion but only 130 evaluable and 79 useful motifs, with 43.79 median effective topics;
- fragment/loss balancing raised optimized coverage from 609 to 911 but did not repair evaluable/useful breadth or SOS;
- post-hoc temperature sharpening changed mixture scale but could not jointly preserve breadth, SOS and NLL;

- pooled projected models contained a 614-topic near-exact beta component;
- entmax without routing produced exact sparsity by starving the inventory;
- the maintained ECRTM Sinkhorn solver failed its convergence contract at real $K = 1000$ and vocabulary 21,233, so no partial model was scored; and
- a reference short-document NSTM failed the paired high-K synthetic screen and was not promoted to MSnLib.

These results justify the selected posterior adaptation without claiming that every future topic model must fail. They also show why another likelihood-only comparison would be insufficient.

6.1 The simplest coherence add-on failed synthetic promotion

The only post-stability model addition was a train-derived positive-NPMI topic loss with the already-audited weight 1.0 and graph settings. No parameter, decoder, posterior, optimizer schedule or other donor component changed. The term reduced its own graph loss but did not improve planted motif-word recovery.

Table 7: Seed-11 K=36 synthetic triage. Graph loss is lower-is-better; all other cosines are higher-is-better.

Metric	Routing ETM	+ positive-NPMI	Change
Held-out NLL	6.278416	6.287552	+0.009136
True-beta cosine	0.498454	0.491576	-0.006878
True-theta cosine	0.764875	0.765219	+0.000344
Train-graph loss	5.526062	5.499502	-0.026559
Active topics	14	14	0
Unique top-1 topics	14	14	0
Median effective topics	1.971	1.955	-0.017

The predeclared gate required at least +0.01 true-beta cosine. The observed change was -0.006878, while NLL worsened by 0.15%. Seeds 23/37, K=128 and real MSⁿLib were therefore not run. This result does not prove that every NPMI coefficient must fail; it shows that coefficient search is not justified by the simplest fixed intervention.

6.2 Removing the one-scalar context failed non-inferiority

The final simplification test retained top-2 token-to-topic voting and entmax but removed the leave-one-out spectral context and its learned scalar. The candidate therefore had exactly the balanced-ETM state and parameter count. The protocol and non-inferiority thresholds were committed before implementation or training.

Table 8: Seed-11 K=36 zero-parameter triage. The token-only route improves over entmax alone but does not retain the selected contextual model.

Metric	Entmax ETM	Top-2 token	Top-2 context
Held-out NLL	6.616673	6.344000	6.278416
True-beta cosine	0.250317	0.410354	0.498454
True-theta cosine	0.403055	0.661425	0.764875
Recovered planted motifs	2	6	10
Active topics	5	11	14
Unique top-1 topics	4	10	14
Median effective topics	1.816	2.005	1.971
Learned parameters	2,167,400	2,167,400	2,167,401

Against the contextual reference, beta and theta fell by 0.088100 and 0.103449, well beyond the allowed 0.02. NLL was 1.0446% worse, recovered motifs fell from 10 to 6, and active/unique topics fell by 3/4. The candidate stayed finite, sparse and free of catastrophic duplication, but failed the model’s central recovery and inventory purpose. Seeds 23/37, K=128 and real MSⁿLib were therefore not run. Top-2 restriction explains part of the repair over entmax alone; leave-one-out spectrum context supplies an additional essential signal.

7 Runtime and system load

The original frozen model trained for 120 epochs in 876.8 seconds (14.6 minutes) on an NVIDIA GeForce RTX 5070. The two added seeds trained in 795.4 and 798.5 seconds (about 13.3 minutes each). Including final inference and evidence hashing, each run completed in roughly 15–16 minutes. Observed GPU load ranged

from moderate to high; one repeat snapshot recorded 40% utilization, 54 °C, 110 W of a 250 W limit and 2.6/12.2 GB device memory in use.

Peak PyTorch allocated/reserved CUDA memory was 0.821/1.053 GB. Process high-water memory was 2.879 GB, with at least 12.43 GB system memory remaining. Deterministic full-validation inference processed approximately 23,426 spectra per second. The load was sustained but comfortably below GPU memory, thermal, power and system memory limits.

8 Limitations and robustness

The three real runs establish descriptive initialization stability on one positive-mode data split: the sample is small and intentionally changes neither data membership nor hyperparameters. It does not establish generalization to other splits, instruments, ionization modes, biological samples, datasets or topic counts.

The positive-NPMI stopping result is one synthetic seed because its predeclared first-stage gate failed. It supports rejection of this fixed add-on, not a broad claim that all possible coherence objectives are ineffective.

The zero-parameter routing result is likewise one synthetic seed because four primary non-inferiority conditions failed at the first stage. It supports retaining the contextual scalar; it is not a claim that no future encoder could use a different parameter-free context representation.

MAG/SOS is a library-dependent chemical proxy. Higher evaluable breadth can lower an unweighted mean by admitting additional borderline topics, which is why counts, mean, median and SOS bands are reported together. Expert interpretation and independent-dataset replication remain necessary.

M1 is retained as a locked private comparator and donor-ablation source, not as the proposed publication architecture. Tomotopy remains the independent production comparator. No claim is made that Routing ETM should replace the production backend before independent and untouched-test review.

9 Recommended next evidence

The baseline and stability checkpoints should be preserved before further model changes. The next steps are:

1. verify both committed source/evidence packages and retained local artifacts;
2. stop repeating identical same-split seeds; future robustness work should vary the split or dataset;
3. do not tune the failed NPMI coefficient or add another private-donor component;
4. retain the one-scalar leave-one-out context rather than replacing it with the failed token-only route;
5. prefer a different split or external library over further architecture search on the same validation evidence; and
6. unlock candidate test only after an independent review freezes the method, checkpoint and success interpretation.

Do not add M1's document gate, Sinkhorn balancing, prototype separation, alternating optimizer or temperature schedule. Additional complexity now risks making an already strong and explainable result harder to publish.

10 Conclusion

Routing-informed sparse ETM repairs the central failure of neural topic models on short MS/MS documents: it achieves sparse per-spectrum mixtures without collapsing or starving the global topic inventory. It exceeds both M1 and Tomotopy on evaluable and useful validation motif counts, while retaining a recognizable ETM generator and adding only one learned scalar.

The result is not uniform dominance. M1 retains better optimized coverage, mean SOS and completion NLL; Tomotopy has higher SOS over fewer evaluable motifs. The predeclared all-gates Boolean remains false and candidate test remains locked. Nevertheless, the scientific outcome is positive: Routing ETM is the strongest paper-facing discovery baseline produced by this campaign, and its central advantage is stable across the three tested initializations. Future work should preserve explainability and test new sources of generalization. The failed NPMI triage reinforces that the simplest supported formulation is the existing Routing ETM, not Routing ETM plus additional donor mechanisms. The failed zero-parameter ablation closes the opposite direction as well: removing leave-one-out context loses too much recovery and inventory for the saving of a single scalar.

A Reproduction and provenance

The frozen implementation and result package were first committed at 3d9af67; the full source commit is recorded in the checkpoint manifest. Committed evidence includes synthetic summaries, the K=128 stress, real

config and metrics, per-topic chemical scores, training history, theta support, duplicate-component diagnostics, fragment/loss balance, top words, environment, validation-access audit and provenance.

Verify the committed checkpoint without training:

```
conda run -n ms2lda-neural python \
  -m scripts.verify_routing_etm_checkpoint
conda run -n ms2lda-neural python \
  -m scripts.verify_routing_etm_stability
conda run -n ms2lda-neural python \
  -m scripts.verify_routing_etm_npmi --require-external
conda run -n ms2lda-neural python \
  -m scripts.verify_routing_etm_top2_token --require-external
```

On the original host, verify every immutable validation input and retained large model artifact:

```
conda run -n ms2lda-neural python \
  -m scripts.verify_routing_etm_checkpoint \
  --verify-inputs --verify-local-artifacts --require-external
conda run -n ms2lda-neural python \
  -m scripts.verify_routing_etm_stability \
  --verify-inputs --verify-local-artifacts --require-external
```

The full local checks complete 78 baseline, 117 stability, 26 stopped-NPMI and 52 zero-parameter-ablation integrity and cross-file consistency checks. Exact replay commands and compact package locations are indexed in `docs/research/README.md`. Each main checkpoint has a machine-readable `checkpoint_manifest.json`. Large weights, NumPy arrays, FAISS indexes, databases and raw MSnLib assets are retained outside Git and identified by path, byte size and SHA-256.

B Frozen diagnostic settings

The reporting contract uses mean topic usage 0.0005, duplicate-component cosine thresholds 0.95/0.99/0.999, top 20 words, catastrophic strict-component fraction 0.5, fragment-mass extremes below 0.1 or above 0.9, and spectrum membership threshold 0.5. These are frozen reporting and failure-detection settings.

References

- [1] J. J. J. van der Hooft et al. “Topic modeling for untargeted substructure exploration in metabolomics.” *Proceedings of the National Academy of Sciences* 113, 13738–13743 (2016). <https://doi.org/10.1073/pnas.1608041113>
- [2] M. Torres Ortega et al. “Large-scale discovery and annotation of substructure patterns in mass spectrometry profiles.” *Nature Communications* 17, 8350 (2026). <https://doi.org/10.1038/s41467-026-75038-0>
- [3] C. Brungs et al. “MSⁿLib: efficient generation of open multi-stage fragmentation mass spectral libraries.” *Nature Methods* 22, 2028–2031 (2025). <https://doi.org/10.1038/s41592-025-02813-0>
- [4] A. B. Dieng, F. J. R. Ruiz, and D. M. Blei. “Topic Modeling in Embedding Spaces.” *Transactions of the Association for Computational Linguistics* 8, 439–453 (2020). https://doi.org/10.1162/tacl_a_00325
- [5] B. Peters, V. Niculae, and A. F. T. Martins. “Sparse Sequence-to-Sequence Models.” *Proceedings of ACL 2019*, 1504–1519 (2019). <https://doi.org/10.18653/v1/P19-1146>
- [6] X. Wu, X. Dong, T. Nguyen, and A. T. Luu. “Effective Neural Topic Modeling with Embedding Clustering Regularization.” *Proceedings of the 40th International Conference on Machine Learning*, 2023. <https://proceedings.mlr.press/v202/wu23c.html>
- [7] J. Dietrich, J. Wandy, L. R. Torres Ortega, and J. J. J. van der Hooft. *vdhooftcompmet/MS2LDA: Paper—Data and Models*. Zenodo (2026). <https://doi.org/10.5281/zenodo.20179680>